

Fig. 7: Block diagram of an embedded navigation system



Fig. 8: An embedded rain-sensing system

overloading any of the four wheels and causing a slip, through either throttle or brake application. When the drive wheels start losing traction, the dynamic stability control automatically begins stabilisation measures. The dynamic stability control system curbs engine output and stops slips on the wheels. In exceptional situations, however, a small amount of wheel slip can be an advantage.

The principle of the traction control system is the adaptation of wheel torque to the coefficient of friction between the wheel and the road surface.

When the traction control computer detects



Fig. 9: Parking spotted by driver with distance mentioned between vehicle and parked vehicle



Fig. 10: Automatic parking using sensors

one or more driven wheels spinning significantly faster than another, it invokes the anti-lock braking system's electronic control unit to apply brake friction to wheels spinning with lessened traction. Braking action on slipping wheel(s) cause power transfer to wheel axle(s) with traction due to the mechanical action within the differential. All-wheel drive vehicles often have an electronically-controlled coupling system in the transfer case to supply non-slipping wheels with torque.

A traction control system has three main components: a sensor equipped in each wheel that senses changes in speed due to traction, an electronic control unit that receives the sensed speed from each wheel and an automatic traction control valve that supports in braking after the electronic control unit processes information from the wheels.

Embedded navigation system

The navigation system consists of an embedded circuitry built with a GPS receiver, gyroscope, DVD-ROM and display system. The GPS receiver receives current longitude and latitude values that are compared with the stored map. The gyroscope and other sensors provide road direction and speed. From all the information gathered at the main controller, the display system displays a navigation or route map of the destination on the display screen.

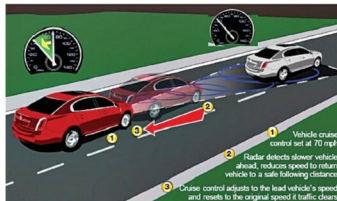


Fig. 11: Cruise control